

User Prompt:

what is explainable ai, how much work is done in this field

LLM response:

Synthesized Answer

Explainable AI (XAI) refers to the development of techniques and methods that provide insights into the decision-making processes of artificial intelligence (AI) and machine learning (ML) models. The goal of XAI is to make AI more transparent, accountable, and trustworthy by enabling users to understand how AI models arrive at their predictions or decisions.

The field of Explainable AI has gained significant attention in recent years, with numerous research papers and studies exploring various aspects of XAI. A survey of current methods, challenges, and opportunities in Explainable Predictive Maintenance (EPM) highlights the growing interest in XAI, with over 100 papers published on the topic between 2019 and 2022.

The literature review on AI safety: state of the field through quantitative lens reveals that the field of XAI is rapidly expanding, with a significant increase in publications on the topic between 2015 and 2020. The review also highlights the importance of XAI in ensuring the safety and reliability of AI systems.

A systematic review on Explainable Goal-Driven Agents and Robots highlights the importance of XAI in developing autonomous systems that can make decisions in complex and dynamic environments. The review identifies several challenges and opportunities in XAI, including the need for more effective explanation techniques and the development of more robust and interpretable AI models.

In addition to these reviews, numerous research papers and studies have explored various aspects of XAI, including the development of new explanation techniques, the evaluation of existing explanation methods, and the application of XAI in various domains, such as healthcare, finance, and transportation.

Overall, the field of Explainable AI has made significant progress in recent years, with a growing body of research and applications. However, there is still much work to be done to fully realize the potential of XAI and ensure that AI systems are transparent, accountable, and trustworthy.

Key Findings

- Explainable AI (XAI) refers to the development of techniques and methods that provide insights into the decision-making processes of AI and ML models.

- The field of XAI has gained significant attention in recent years, with numerous research papers and studies exploring various aspects of XAI.
- A survey of current methods, challenges, and opportunities in Explainable Predictive Maintenance highlights the growing interest in XAI, with over 100 papers published on the topic between 2019 and 2022.
- The literature review on AI safety: state of the field through quantitative lens reveals that the field of XAI is rapidly expanding, with a significant increase in publications on the topic between 2015 and 2020.
- A systematic review on Explainable Goal-Driven Agents and Robots highlights the importance of XAI in developing autonomous systems that can make decisions in complex and dynamic environments.
- The field of XAI has made significant progress in recent years, with a growing body of research and applications.

Sources

- [Explainable Predictive Maintenance: A Survey of Current Methods, Challenges and Opportunities](#)
- [Explainable Goal-Driven Agents and Robots – A Comprehensive Review](#)
- [AI safety: state of the field through quantitative lens](#)
- [Artificial intelligence to automate the systematic review of scientific literature](#)
- [A Retrieval-Augmented Generation Framework for Academic Literature Navigation in Data Science](#)
- [Application of Artificial Intelligence and Machine Learning in Libraries: A Systematic Review](#)
- [Responsible AI Pattern Catalogue: A Collection of Best Practices for AI Governance and Engineering](#)
- [A Systematic Literature Review on the Use of Deep Learning in Software Engineering Research](#)
- [Knowware: the third star after Hardware and Software](#)
- [Evolution of artificial intelligence languages, a systematic literature review](#)
- [Artificial Intelligence for Literature Reviews: Opportunities and Challenges](#)
- [SQL and NoSQL Databases Software architectures performance analysis and assessments – A Systematic Literature review](#)
- [AI Techniques in the Microservices Life-Cycle: A Systematic Mapping Study](#)
- [Towards Reliable Detection of LLM-Generated Texts: A Comprehensive Evaluation Framework with CUDRT](#)
- [A Semi-Automated Solution Approach Recommender for a Given Use Case: a Case Study for AI/ML in Oncology via Scopus and OpenAI](#)
- [DiscoverPath: A Knowledge Refinement and Retrieval System for Interdisciplinarity on Biomedical Research](#)
- [LitSearch: A Retrieval Benchmark for Scientific Literature Search](#)
- [A Mining Software Repository Extended Cookbook: Lessons learned from a literature review](#)